

Exploratory Data Analysis

(EDA)

A Structured Theory & Practice Guide
For Big Data Engineering, SQL & PySpark Interviews

Data Quality

Statistics

Distributions

Outliers

Correlation

Time Series

SQL

PySpark

"What is inside my data, and can I trust it?"

Introduction — Why EDA Matters

EDA (Exploratory Data Analysis) is the process of examining, cleaning, summarizing, and visualizing data to understand its structure, quality, patterns, relationships, and problems — **before** any serious analysis, machine learning, dashboarding, or business decision-making happens.

● **Key question:** *What is inside my data, and can I trust this data for further analysis?*

The EDA Flow

Raw Data

- |
1. Understand the Dataset
2. Inspect Schema & Data Types
3. Check Data Quality
4. Handle Missing Values
5. Detect Duplicates
6. Detect Invalid Values
7. Analyze Distributions
8. Analyze Categorical Data
9. Analyze Relationships
10. Detect Outliers
11. Analyze Trends
12. Generate Insights

Clean / Trusted Dataset

01 Understand the Business Problem

The first and most important step. Before looking at columns, ask: **why does this dataset exist?**

Example — Sales Dataset

order_id	date	customer	product	category	quantity	price
1001	2026-01-01	Ahmed	Laptop	Electronics	1	1000
1002	2026-01-02	Ali	Mouse	Electronics	3	20

You need to determine:

- What does one row represent?
- What is the business process?
- What does each column mean?
- What questions are we trying to answer?

Grain / Granularity

Grain is **extremely important**. Many EDA mistakes happen because the analyst doesn't understand it.

If 1 row =	Then COUNT(*) means
1 transaction	Number of transactions
1 customer	Number of customers

02 Understand the Dataset Structure

Inspect: how many rows, how many columns, column names, data types, and what each column represents. Then classify every column into a category:

Column type	Examples	Notes
Numerical	age, salary, quantity, price, revenue	Support math operations (mean, sum...)
Categorical	country, city, gender, category	Represent groups
Date/time	order_date, created_at, timestamp	Enable trend/seasonality analysis
Identifier	customer_id, order_id, product_id	Numeric-looking but not analytical — never average an ID

03 Inspect the Schema

Ask: **are the data types correct?**

Column	Current type	Should probably be
order_date	STRING	DATE
price	STRING	DOUBLE / DECIMAL

If price is stored as text ("100", "200"), you cannot reliably perform numerical calculations until it is converted.

04 Understand Data Quality

- **Key question:** *Is the data clean and trustworthy?*

Common data-quality problems to look for:

- Missing values
- Duplicates
- Invalid values
- Wrong data types
- Inconsistent formats
- Outliers
- Impossible values
- Inconsistent categories

05 Missing Value Analysis

A missing value means information is unavailable.

customer	age	city
Ahmed	25	Cairo
Ali	NULL	Giza
Sara	30	NULL

Determine: which columns have NULLs, how many, what percentage, why they're missing, and what to do about it.

Counts alone aren't enough — 100 NULLs out of 1,000,000 rows (0.01%) is very different from 100 out of 500 (20%).

06 Handling Missing Values

There isn't one universal solution:

Strategy	When to use
Remove rows	Only a tiny number of rows affected
Replace with mean	Numerical, roughly symmetric distribution
Replace with median	Numerical, skewed distribution (more robust)
Replace with mode	Categorical — use most common value
Use a special category	e.g. city = NULL → "Unknown"
Keep NULL	NULL itself carries meaning, e.g. cancellation_date = NULL means not cancelled

Don't blindly replace every NULL — understand what it means first.

07 Duplicate Analysis

Is a repeated `order_id` a real duplicate, or does an order legitimately contain multiple products? This is why **grain** must be understood first.

```
Total rows
|
Distinct rows
|
Duplicate rows
```

Also check duplicates against business keys: `order_id` may need to be unique, but `customer_id` usually doesn't in a transaction table.

08 Analyze Unique Values

For categorical columns, examine DISTINCT values — this reveals data-quality issues like inconsistent categories:

```
gender :  
Male  
Female  
Unknown  
M  
F  
male  
female
```

Male, M, and male likely represent the same category and need standardization.

09 Frequency Analysis

Country	Count
Egypt	50,000
USA	30,000
Germany	15,000
France	5,000

Common questions: most/least common category, rare categories, is the data heavily imbalanced?

10 Numerical Statistical Analysis

For numerical columns, calculate descriptive statistics:

Statistic	Salary example
Count	10,000
Mean	8,500
Median	7,000
Minimum	2,000
Maximum	100,000
Std Dev	9,500

11 Mean

Mean = sum of values ÷ number of values. It can be heavily skewed by extreme values:

Dataset	Values	Mean
Normal	10, 20, 30, 40, 50	30
With outlier	10, 20, 30, 40, 1000	220 (misleading — most values are 10–40)

12 Median

The middle value after sorting. For 10, 20, 30, 40, 1000 the median is still **30** — **median is more robust against outliers** than mean.

13 Standard Deviation

Spread	Example values
Low std dev	100, 101, 102, 103, 104 (close together)
High std dev	10, 50, 100, 500, 1000 (widely spread)

Think: **Mean = center**, **Standard deviation = spread**.

14 Percentiles & IQR

Percentile	Value
25th (Q1)	2,000
50th (Median)	5,000
75th (Q3)	8,000

$IQR = Q3 - Q1$
(commonly used for outlier detection)

15 Distribution Analysis

● **Key question:** *How are numerical values distributed?*

- Normal / symmetric
- Right-skewed
- Left-skewed
- Uniform
- Bimodal

Example: most employees earn 5,000–10,000 while a few earn 50,000–100,000 → the distribution is **right-skewed**. Understanding shape helps choose the right statistical methods.

16 Outlier Detection

An outlier is a value unusually far from the rest of the data. **An outlier isn't automatically an error** — a \$500,000 salary might be legitimate for a CEO. Ask: is this an error, or a legitimate extreme value?

17 IQR Method for Outliers

$IQR = Q3 - Q1$

Lower Bound = $Q1 - 1.5 \times IQR$

Upper Bound = $Q3 + 1.5 \times IQR$

Example:

$Q1 = 20, Q3 = 40 \rightarrow IQR = 20$

Lower = $20 - 1.5(20) = -10$

Upper = $40 + 1.5(20) = 70$

Values above 70 (or below -10) are potential outliers.

18 Relationships Between Variables

Instead of analyzing single columns, examine pairs: salary↔experience, price↔quantity, sales↔month, age↔spending.

19 Correlation

Correlation value	Meaning
0.85	Strong positive — as experience increases, salary tends to increase
-0.80	Strong negative — as one increases, the other tends to decrease
≈ 0	Little linear relationship

Interview point: Correlation does NOT imply causation. Ice cream sales and drowning incidents both rise together, but ice cream doesn't cause drowning — a third variable (temperature) drives both.

20 Categorical vs Numerical Analysis

Compare a numerical variable across categories, e.g. average salary by department:

Department	Avg Salary
IT	12,000
Finance	10,000
Marketing	9,000
HR	8,000

21 Group-Based Analysis

```
GROUP BY category  
|  
COUNT / SUM / AVG / MIN / MAX
```

Category	Total Sales
Electronics	500,000
Furniture	300,000
Clothing	200,000

EDA shouldn't stop at the numbers — always ask *why* a category is highest or lowest.

22 Time-Series Analysis

- Are sales increasing or decreasing?
- Are there seasonal patterns?
- Are there unusual spikes or drops?

Month	Sales
January	100,000
February	110,000
March	115,000
April	130,000
...	...
December	250,000

23 Trend vs Seasonality

Concept	Definition	Example
Trend	Long-term direction	2023: 100K → 2026: 180K, overall increasing
Seasonality	Repeated pattern over a fixed period	Sales spike every December, every year

24 Data Consistency (Business Rules)

Check whether values follow expected business rules — likely invalid examples:

- `age < 0`
- `quantity = -10`
- `discount = 150%`
- `order_date > current_date`

25 Referential Integrity

Across related tables (customers, orders, products), check that foreign keys resolve — e.g. `orders.customer_id` should exist in `customers.customer_id`. An `orders.customer_id = 99999` with no matching customer is an **orphan record** — especially important in data engineering.

26 EDA With SQL

```
SELECT *  
FROM table  
LIMIT 10;  
  
-- then:  
COUNT(*)  
COUNT(DISTINCT ...)  
MIN() / MAX() / AVG() / SUM()  
  
-- then:  
GROUP BY / ORDER BY / HAVING / JOIN
```

Example query

```
SELECT  
    category,  
    COUNT(*)           AS total_orders,  
    SUM(quantity * price) AS total_sales,  
    AVG(price)         AS avg_price  
FROM sales  
GROUP BY category  
ORDER BY total_sales DESC;
```

Answers: which category generates the most sales, how many orders per category, and what's the average price.

27 EDA With PySpark

PySpark is ideal for EDA when the dataset is too large for Pandas.

Data Source -> Spark DataFrame -> Schema Inspection -> Data Quality
-> Descriptive Statistics -> Aggregations -> Group Analysis
-> Relationships -> Insights

```
df.printSchema()  
df.show(10)  
df.describe().show()  
df.select("category").distinct().show()  
df.groupBy("category").count().show()
```

```
df.groupBy("category").agg(  
    sum("sales").alias("total_sales"),  
    avg("sales").alias("average_sales"),  
    max("sales").alias("maximum_sales")  
).show()
```

28 Why PySpark EDA Matters at Scale

A 500 GB CSV won't comfortably fit into Pandas on a normal machine:

HDFS / Cloud Storage / Data Lake

|

Spark

|

Distributed Processing

|

EDA Results

Pandas EDA

|

PySpark EDA

|

Same analytical questions

|

Different execution engine

29 Visualization

Chart	Best for
Histogram	Distribution of age, salary, price
Box plot	Outliers, median, quartiles
Bar chart	Sales by category, customers by country
Line chart	Sales/revenue over time
Scatter plot	Salary vs experience, price vs quantity
Heatmap	Correlation matrix

In Big Data, aggregate with Spark first, *then* visualize the smaller result — don't collect hundreds of GB to the driver just to make a chart:

500 GB raw data -> Spark aggregation -> 500 rows -> Visualization

30 EDA Is Not Just Cleaning

Interview point: EDA is not "remove NULLs + remove duplicates." It includes understanding, data quality, statistics, distribution, relationships, patterns, outliers, trends, and business insights. Cleaning is one component, not the whole process.

31 The Complete EDA Framework



32 Real-World Walkthrough — sales.csv

Columns: order_id, order_date, customer_id, country, category, product, quantity, unit_price, discount, payment_method.

Step	What to do
1. Understand	One row = one transaction/product sale
2. Schema	order_id int/str, order_date date, quantity int, unit_price decimal
3. Data quality	NULLs, duplicates, invalid dates, negative quantities/prices, invalid discounts
4. Statistics	mean/median/min/max/std/percentiles for quantity, unit_price, discount
5. Categories	country, category, payment_method
6. Group analysis	sales by country / category / product / month
7. Outliers	unusually large transactions, prices, quantities
8. Relationships	quantity vs sales, discount vs sales, price vs quantity
9. Time analysis	daily / monthly / yearly sales
10. Insights	translate findings into business knowledge

Example Insights Produced

- Electronics generates 45% of revenue.
- Egypt represents the largest number of orders.
- Sales increase significantly in December.
- 5% of transactions have missing payment methods.
- A small number of transactions contain unusually high quantities.
- Discounts appear to be associated with higher quantities.

This is what turns EDA output into **business knowledge**, not just tables.

33 EDA Deliverables Checklist

- Dataset overview
- Data dictionary
- Data quality report
- Missing-value analysis
- Duplicate analysis
- Descriptive statistics
- Categorical analysis
- Numerical analysis
- Distribution analysis
- Outlier analysis
- Correlation / relationship analysis
- Time-series analysis
- Visualizations
- Business insights
- Data-cleaning decisions

34 EDA vs Data Cleaning vs Data Analysis

Activity	Focus	Examples
Data Cleaning	Fixing data problems	NULL handling, dedup, type correction, standardization
EDA	Understanding the data	Distributions, patterns, outliers, relationships, trends, quality
Data Analysis	Answering business questions	"Why did revenue decrease in Q2?"

Data Cleaning -> EDA -> Data Analysis -> Business Decisions

In real projects these activities overlap and iterate rather than happening in a strictly linear sequence.

35 EDA in a Big Data Engineering Pipeline



This is where EDA connects directly to the medallion-architecture pipeline: bronze → silver → gold.

The Most Important Idea to Remember

When given a new dataset, **don't immediately start writing transformations**. Think through these ten questions first:

- What is this data?
- What does one row represent?
- What does every column mean?
- Can I trust the data?
- What is missing or incorrect?
- How is the data distributed?
- Are there outliers?
- Are there relationships between variables?
- Are there trends over time?
- What business insights can I extract?

For a Big Data interview: be ready to explain EDA using both SQL and PySpark. The analytical logic is the same in both — the difference is that PySpark performs these operations in a **distributed** environment on large-scale datasets.